

Linear Equations & Systems

ANSWER KEY



Solving single-variable linear equations

- 1 Solve for x : $3x - 7 = 20$.

$$3x = 27, \text{ so } x = 9.$$

- 2 Solve for x : $\frac{2x + 5}{3} = 7$.

Multiply both sides by 3: $2x + 5 = 21$, so $2x = 16$, giving $x = 8$.

- 3 Solve for x : $5(x - 2) = 3(x + 4)$.

Expand: $5x - 10 = 3x + 12$. Subtract $3x$: $2x - 10 = 12$, so $2x = 22$, giving $x = 11$.

Equations with no solution or infinitely many solutions

- 4 For what value of k does the equation $4x + 6 = kx + 6$ have infinitely many solutions?

For infinitely many solutions, both sides must be identical for every x , so the coefficients of x must match: $k = 4$ (and the constants already match at $6 = 6$).

- 5 Explain why the equation $2x + 3 = 2x + 7$ has no solution, without solving it algebraically first.

Both sides have the same coefficient of x (2), so subtracting $2x$ from both sides leaves $3 = 7$, a false statement independent of x . Since no value of x can make a false numerical statement true, the equation has no solution.

Solving systems by substitution

- 6 Solve the system: $y = 2x + 1$ and $3x + y = 16$.

Substitute: $3x + (2x + 1) = 16$, so $5x + 1 = 16$, giving $5x = 15$, $x = 3$. Then $y = 2(3) + 1 = 7$. Solution: $(3, 7)$.

- 7 Solve the system: $x - 2y = 5$ and $2x + 3y = 3$.

From the first equation, $x = 5 + 2y$. Substitute into the second: $2(5 + 2y) + 3y = 3$, so $10 + 4y + 3y = 3$, giving $7y = -7$, $y = -1$. Then $x = 5 + 2(-1) = 3$. Solution: $(3, -1)$.

Solving systems by elimination

- 8 Solve the system: $2x + y = 11$ and $x - y = 1$, using elimination.

Adding the two equations directly eliminates y : $3x = 12$, so $x = 4$. Substituting into $x - y = 1$: $4 - y = 1$, so $y = 3$. Solution: $(4, 3)$.

- 9 Solve the system: $3x + 2y = 16$ and $5x - 2y = 8$, using elimination.

Adding the two equations eliminates y : $8x = 24$, so $x = 3$. Substituting into $3x + 2y = 16$: $9 + 2y = 16$, so $2y = 7$, giving $y = 3.5$. Solution: $(3, 3.5)$.

Systems with no solution or infinitely many solutions

- 10 For what value of c does the system $6x + 9y = 12$ and $2x + 3y = c$ have infinitely many solutions?

Dividing the first equation by 3 gives $2x + 3y = 4$. For the system to have infinitely many solutions, the second equation must represent the same line, so $c = 4$.

- 11 Explain why the system $4x - 2y = 10$ and $2x - y = 3$ has no solution.

Dividing the first equation by 2 gives $2x - y = 5$. This directly contradicts the second equation, $2x - y = 3$ — the same left-hand expression cannot equal two different values simultaneously, so the two lines are parallel (same slope, different intercepts) and never intersect.

Translating word problems into linear equations

- 12 A rectangle's length is 4 more than twice its width. If the perimeter is 68, find the width and length.

Let width = w , length = $2w + 4$. Perimeter: $2(w) + 2(2w + 4) = 68$, so $2w + 4w + 8 = 68$, giving $6w = 60$, $w = 10$. Length = $2(10) + 4 = 24$.

- 13 A taxi charges a flat fee of 4 plus 2.50 per mile. If a ride costs 21.50, how many miles was the ride?

Let m be the number of miles: $4 + 2.5m = 21.5$, so $2.5m = 17.5$, giving $m = 7$ miles.

Systems as word problems

- 14 A movie theater sells adult tickets for 12 and child tickets for 8. If 150 tickets were sold for a total of 1560, how many of each type were sold?

Let a = adult tickets, c = child tickets. $a + c = 150$ and $12a + 8c = 1560$. From the first equation, $c = 150 - a$. Substitute: $12a + 8(150 - a) = 1560$, so $12a + 1200 - 8a = 1560$, giving $4a = 360$, $a = 90$. Then $c = 150 - 90 = 60$. So 90 adult tickets and 60 child tickets were sold.

- 15 Two numbers have a sum of 45 and a difference of 11. Find the two numbers.

Let the numbers be x and y with $x + y = 45$ and $x - y = 11$. Adding: $2x = 56$, so $x = 28$. Then $y = 45 - 28 = 17$. The numbers are 28 and 17.

Interpreting the slope and intercept of a linear model

- 16 A gym membership costs $C = 25 + 15m$ dollars, where m is the number of months. Interpret the meaning of the numbers 25 and 15 in this context.

The 25 represents a fixed one-time joining fee (the cost when $m = 0$ months), and the 15 represents the monthly membership rate — each additional month increases the total cost by 15 dollars.

- 17 Two companies charge for printing: Company A charges $C = 0.10p + 20$ and Company B charges $C = 0.15p$, where p is the number of pages. Find the number of pages at which both companies charge the same amount, and state which company is cheaper for 500 pages.

Set equal: $0.10p + 20 = 0.15p$, so $20 = 0.05p$, giving $p = 400$ pages. For 500 pages: Company A charges $0.10(500) + 20 = 70$ dollars; Company B charges $0.15(500) = 75$ dollars. Company A is cheaper at 500 pages.

Equations with variables on both sides and fractions

- 18 Solve for x : $\frac{x}{3} + \frac{x}{4} = 7$.

Multiply every term by the least common denominator, 12: $4x + 3x = 84$, so $7x = 84$, giving $x = 12$.

- 19 Solve for x : $2(x + 3) - 5 = 3(x - 1) + 2$.

Expand: $2x + 6 - 5 = 3x - 3 + 2$, so $2x + 1 = 3x - 1$. Subtract $2x$: $1 = x - 1$, giving $x = 2$.

Systems with three unknowns represented as a word problem

- 20 A vending machine sells snacks for 1.50 dollars and drinks for 2 dollars. On one day it sold 45 items total for 79.50 dollars. Set up and solve a system to find the number of snacks and drinks sold.

Let s = snacks, d = drinks: $s + d = 45$ and $1.5s + 2d = 79.5$. From the first, $d = 45 - s$. Substitute: $1.5s + 2(45 - s) = 79.5$, so $1.5s + 90 - 2s = 79.5$, giving $-0.5s = -10.5$, so $s = 21$. Then $d = 45 - 21 = 24$. So 21 snacks and 24 drinks were sold.

Choosing the best method for a given system

- 21 For the system $y = 4x - 3$ and $5x + 2y = 17$, explain which method (substitution or elimination) is more efficient here, and solve the system.

Substitution is more efficient here, since y is already isolated in the first equation. Substitute into the second: $5x + 2(4x - 3) = 17$, so $5x + 8x - 6 = 17$, giving $13x = 23$, $x = \frac{23}{13}$. Then $y = 4\left(\frac{23}{13}\right) - 3 = \frac{92}{13} - \frac{39}{13} = \frac{53}{13}$.

Consistent, inconsistent, and dependent systems

- 22 Classify the system $y = 3x + 2$ and $y = 3x - 4$ as consistent, inconsistent, or dependent, and explain your reasoning without solving.

Both lines have the same slope (3) but different y -intercepts (2 and -4), so they are parallel and never intersect. The system is inconsistent (no solution).

Multiple choice — easy 15

- 23 Solve for x : $4x + 2 = 8x - 58$.

- a. -15
- b. 16
- c. 15 ✓
- d. 14

$$-4x = -60, \text{ so } x = 15.$$

- 24 Solve for x : $-7x + 13 = -8x + 7$.

- a. -6 ✓
- b. -5
- c. 6
- d. -7

$$x = -6.$$

25 Solve for x : $-6x + 10 = 9x - 5$.

a. 1 ✓

b. 2

c. -1

d. 0

$$-15x = -15, \text{ so } x = 1.$$

26 Solve for x : $8x - 10 = 3x - 80$.

a. 14

b. -13

c. -14 ✓

d. -15

$$5x = -70, \text{ so } x = -14.$$

27 Solve for x : $2x + 15 = -8x - 95$.

a. -10

b. -11 ✓

c. -12

d. 11

$$10x = -110, \text{ so } x = -11.$$

28 Solve for x : $9x - 19 = 7x - 37$.

a. 9

b. -10

c. -8

d. -9 ✓

$$2x = -18, \text{ so } x = -9.$$

29 Solve for x : $9x - 19 = -6x - 4$.

a. 1 ✓

b. 2

c. -1

d. 0

$15x = 15$, so $x = 1$.

30 Solve for x : $x - 10 = -8x + 116$.

a. 14 ✓

b. -14

c. 15

d. 13

$9x = 126$, so $x = 14$.

31 Solve for x : $-8x + 14 = x - 13$.

a. 3 ✓

b. -3

c. 2

d. 4

$-9x = -27$, so $x = 3$.

32 Solve for x : $3x - 19 = 8x - 64$.

a. 8

b. 9 ✓

c. -9

d. 10

$-5x = -45$, so $x = 9$.

33 Solve for x : $8x - 4 = -2x - 154$.

a. -15 ✓

b. -14

c. -16

d. 15

$10x = -150$, so $x = -15$.

34 Solve for x : $-8x - 3 = -5x - 24$.

a. 8

b. -7

c. 6

d. 7 ✓

$-3x = -21$, so $x = 7$.

35 Solve for x : $-x + 16 = 3x - 40$.

a. 14 ✓

b. 13

c. 15

d. -14

$-4x = -56$, so $x = 14$.

36 Solve for x : $-3x + 9 = 3x + 75$.

- a. 11
- b. -12
- c. -11 ✓
- d. -10

$-6x = 66$, so $x = -11$.

37 Solve for x : $7x + 1 = -3x + 51$.

- a. 5 ✓
- b. 4
- c. 6
- d. -5

$10x = 50$, so $x = 5$.

Multiple choice — medium 15

38 Solve the system: $-2x + 6y = 22$ and $x + 6y = -2$.

- a. $(-8, 1)$ ✓
- b. $(-8, 2)$
- c. $(-7, 1)$
- d. $(1, -8)$

The solution is $x = -8, y = 1$: $(-8, 1)$.

39 Solve the system: $0x + 3y = -3$ and $5x + 5y = -45$.

a. $(-8, -1)$ ✓

b. $(-1, -8)$

c. $(-8, 0)$

d. $(-7, -1)$

The solution is $x = -8, y = -1$: $(-8, -1)$.

40 Solve the system: $2x - 3y = 11$ and $0x - 6y = -18$.

a. $(10, 4)$

b. $(3, 10)$

c. $(10, 3)$ ✓

d. $(11, 3)$

The solution is $x = 10, y = 3$: $(10, 3)$.

41 Solve the system: $-2x + 5y = -24$ and $-3x - 2y = -17$.

a. $(-2, 7)$

b. $(7, -2)$ ✓

c. $(8, -2)$

d. $(7, -1)$

The solution is $x = 7, y = -2$: $(7, -2)$.

42 Solve the system: $-3x - y = 35$ and $-5x - 3y = 65$.

a. $(-9, -5)$

b. $(-10, -4)$

c. $(-5, -10)$

d. $(-10, -5)$ ✓

The solution is $x = -10, y = -5$: $(-10, -5)$.

43 Solve the system: $-6x - 5y = -4$ and $3x - 2y = 47$.

a. $(9, -9)$

b. $(9, -10)$ ✓

c. $(-10, 9)$

d. $(10, -10)$

The solution is $x = 9, y = -10$: $(9, -10)$.

44 Solve the system: $4x + 5y = 51$ and $-4x - 2y = -30$.

a. $(5, 7)$

b. $(4, 8)$

c. $(7, 4)$

d. $(4, 7)$ ✓

The solution is $x = 4, y = 7$: $(4, 7)$.

45 Solve the system: $-x + 6y = -44$ and $4x + 6y = -34$.

a. $(2, -7)$ ✓

b. $(3, -7)$

c. $(-7, 2)$

d. $(2, -6)$

The solution is $x = 2, y = -7$: $(2, -7)$.

46 Solve the system: $-4x - 2y = -4$ and $-2x - 5y = 38$.

a. $(6, -10)$ ✓

b. $(7, -10)$

c. $(6, -9)$

d. $(-10, 6)$

The solution is $x = 6, y = -10$: $(6, -10)$.

47 Solve the system: $x + 3y = 11$ and $-x + 6y = 34$.

a. $(5, -4)$

b. $(-4, 6)$

c. $(-3, 5)$

d. $(-4, 5)$ ✓

The solution is $x = -4, y = 5$: $(-4, 5)$.

48 Solve the system: $-2x - 6y = 28$ and $2x - 4y = 42$.

a. $(-7, 7)$

b. $(7, -7)$ ✓

c. $(8, -7)$

d. $(7, -6)$

The solution is $x = 7, y = -7$: $(7, -7)$.

49 Solve the system: $0x - y = 3$ and $5x - 4y = 57$.

a. $(9, -3)$ ✓

b. $(-3, 9)$

c. $(9, -2)$

d. $(10, -3)$

The solution is $x = 9, y = -3$: $(9, -3)$.

50 Solve the system: $-4x - 4y = -40$ and $-4x + 4y = 40$.

- a. $(10, 0)$
- b. $(1, 10)$
- c. $(0, 11)$
- d. $(0, 10)$ ✓

The solution is $x = 0, y = 10$: $(0, 10)$.

51 Solve the system: $3x + 3y = -21$ and $-5x - 2y = 35$.

- a. $(-7, 0)$ ✓
- b. $(-7, 1)$
- c. $(-6, 0)$
- d. $(0, -7)$

The solution is $x = -7, y = 0$: $(-7, 0)$.

52 Solve the system: $x + 2y = -12$ and $5x + 6y = -44$.

- a. $(-4, -3)$
- b. $(-4, -4)$ ✓
- c. $(-5, -2)$
- d. $(-3, -4)$

The solution is $x = -4, y = -4$: $(-4, -4)$.

Multiple choice — hard 15

53 For what value of c does the system $10x + 20y = 25$ and $2x + 4y = c$ have infinitely many solutions?

a. 10

b. 5 ✓

c. 0

d. 25

Dividing the first equation by 5 gives $2x + 4y = 5$. For infinitely many solutions, the second equation must represent the same line, so $c = 5$.

54 For what value of c does the system $-15x - 5y = -25$ and $-3x - y = c$ have infinitely many solutions?

a. -5 ✓

b. -10

c. -25

d. 0

Dividing the first equation by 5 gives $-3x - y = -5$. For infinitely many solutions, the second equation must represent the same line, so $c = -5$.

55 For what value of c does the system $-15x - 30y = -60$ and $-3x - 6y = c$ have infinitely many solutions?

a. -17

b. -7

c. -12 ✓

d. -60

Dividing the first equation by 5 gives $-3x - 6y = -12$. For infinitely many solutions, the second equation must represent the same line, so $c = -12$.

56 For what value of c does the system $0x - 6y = -28$ and $0x - 3y = c$ have infinitely many solutions?

- a. -28
- b. -12
- c. -16
- d. -14 ✓

Dividing the first equation by 2 gives $0x - 3y = -14$. For infinitely many solutions, the second equation must represent the same line, so $c = -14$.

57 For what value of c does the system $-8x - 12y = 44$ and $-2x - 3y = c$ have infinitely many solutions?

- a. 7
- b. 15
- c. 44
- d. 11 ✓

Dividing the first equation by 4 gives $-2x - 3y = 11$. For infinitely many solutions, the second equation must represent the same line, so $c = 11$.

58 A store sells swim passes for 20 dollars and gym passes for 12 dollars. If 82 items were sold for a total of 1072 dollars, how many of each type were sold?

- a. (71, 11)
- b. (6, 76)
- c. (11, 71) ✓
- d. (16, 66)

Let a = swim passes, b = gym passes: $a + b = 82$ and $20a + 12b = 1072$. Solving gives $a = 11$ and $b = 71$: (11, 71).

- 59** A store sells burgers for 17 dollars and salads for 13 dollars. If 136 items were sold for a total of 2012 dollars, how many of each type were sold?

- a. (56, 80)
- b. (61, 75) ✓
- c. (75, 61)
- d. (66, 70)

Let a = burgers, b = salads: $a + b = 136$ and $17a + 13b = 2012$. Solving gives $a = 61$ and $b = 75$: (61, 75).

- 60** A store sells burgers for 8 dollars and salads for 3 dollars. If 85 items were sold for a total of 305 dollars, how many of each type were sold?

- a. (75, 10)
- b. (15, 70)
- c. (10, 75) ✓
- d. (5, 80)

Let a = burgers, b = salads: $a + b = 85$ and $8a + 3b = 305$. Solving gives $a = 10$ and $b = 75$: (10, 75).

- 61** A store sells adult tickets for 9 dollars and child tickets for 3 dollars. If 54 items were sold for a total of 288 dollars, how many of each type were sold?

- a. (33, 21)
- b. (26, 28)
- c. (16, 38)
- d. (21, 33) ✓

Let a = adult tickets, b = child tickets: $a + b = 54$ and $9a + 3b = 288$. Solving gives $a = 21$ and $b = 33$: (21, 33).

- 62 A store sells burgers for 6 dollars and salads for 2 dollars. If 79 items were sold for a total of 294 dollars, how many of each type were sold?

- a. (29, 50)
- b. (34, 45) ✓
- c. (39, 40)
- d. (45, 34)

Let a = burgers, b = salads: $a + b = 79$ and $6a + 2b = 294$. Solving gives $a = 34$ and $b = 45$: (34, 45).

- 63 Solve for x : $x/4 + x/6 = 20$.

- a. 60
- b. 48 ✓
- c. 36
- d. 20

Multiply every term by the least common denominator, 12: $3x + 2x = 240$, so $5x = 240$, giving $x = 48$.

- 64 Solve for x : $x/3 + x/5 = 8$.

- a. 15 ✓
- b. 8
- c. 1
- d. 30

Multiply every term by the least common denominator, 15: $5x + 3x = 120$, so $8x = 120$, giving $x = 15$.

65 Solve for x : $x/3 + x/5 = 40$.

- a. 60
- b. 40
- c. 75 ✓
- d. 90

Multiply every term by the least common denominator, 15: $5x + 3x = 600$, so $8x = 600$, giving $x = 75$.

66 Solve for x : $x/5 + x/6 = 55$.

- a. 55
- b. 150 ✓
- c. 120
- d. 180

Multiply every term by the least common denominator, 30: $6x + 5x = 1650$, so $11x = 1650$, giving $x = 150$.

67 Solve for x : $x/2 + x/4 = 6$.

- a. 4
- b. 6
- c. 8 ✓
- d. 12

Multiply every term by the least common denominator, 4: $2x + 1x = 24$, so $3x = 24$, giving $x = 8$.